University of Ottawa

Department of Mathematics and Statistics

MAT 2362 : Foundations of Mathematics

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Midterm Exam

November 3, 2023

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) All work to be considered for grading should be written on your solution sheets. You should clearly indicate where the solution of each question is written.
(c) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(d) You are strongly recommended to write in **pen**, not pencil.

Please do not write in the table below.

Question	1	2	3	4	5	Total
Maximum	3	5	5	2	4	19
Grade						

1. **[3 points]** Prove the following equivalence using algebraic equivalences. You should *not* use truth tables. Justify each step by writing the number of the equivalence you are using from the table below.

$$(X \Rightarrow Y) \vee (X \wedge Y) \equiv (X \Rightarrow Y)$$

Solution:

$$\begin{aligned}
(X \Rightarrow Y) \vee (X \wedge Y) &\equiv (Y \vee \neg X) \vee (X \wedge Y) && (18)\\
&\equiv (Y \vee \neg X \vee X) \wedge (Y \vee \neg X \vee Y) && (17), (3)\\
&\equiv (Y \vee \mathbf{T}) \wedge (Y \vee \neg X \vee Y) && (12)\\
&\equiv \mathbf{T} \wedge (Y \vee Y \vee \neg X) && (9), (1)\\
&\equiv Y \vee Y \vee \neg X && (8)\\
&\equiv Y \vee \neg X && (5)\\
&\equiv X \Rightarrow Y && (18).
\end{aligned}$$

	Equivalence			Name or comment
(1)	$P \vee Q$	$\equiv$	$Q \vee P$	Commutativity of $\vee$
(2)	$P \wedge Q$	$\equiv$	$Q \wedge P$	Commutativity of $\wedge$
(3)	$(P \vee Q) \vee R$	$\equiv$	$P \vee (Q \vee R)$	Associativity of $\vee$
(4)	$(P \wedge Q) \wedge R$	$\equiv$	$P \wedge (Q \wedge R)$	Associativity of $\wedge$
(5)	$P \vee P$	$\equiv$	P	Idempotence
(6)	$P \wedge P$	$\equiv$	P	Idempotence
(7)	$P \vee \mathbf{F}$	$\equiv$	P	$\mathbf{F}$ is neutral for $\vee$
(8)	$P \wedge \mathbf{T}$	$\equiv$	P	$\mathbf{T}$ is neutral for $\wedge$
(9)	$P \vee \mathbf{T}$	$\equiv$	$\mathbf{T}$	$\mathbf{T}$ is absorbent for $\vee$
(10)	$P \wedge \mathbf{F}$	$\equiv$	$\mathbf{F}$	$\mathbf{F}$ is absorbent for $\wedge$
(11)	$\neg\neg P$	$\equiv$	P	Double negation
(12)	$P \vee \neg P$	$\equiv$	$\mathbf{T}$	Excluded middle
(13)	$P \wedge \neg P$	$\equiv$	$\mathbf{F}$	Contradiction
(14)	$\neg(P \vee Q)$	$\equiv$	$\neg P \wedge \neg Q$	De Morgan's law
(15)	$\neg(P \wedge Q)$	$\equiv$	$\neg P \vee \neg Q$	De Morgan's law
(16)	$P \wedge (Q \vee R)$	$\equiv$	$(P \wedge Q) \vee (P \wedge R)$	Distributivity of $\wedge$ over $\vee$
(17)	$P \vee (Q \wedge R)$	$\equiv$	$(P \vee Q) \wedge (P \vee R)$	Distributivity of $\wedge$ over $\vee$
(18)	$P \Rightarrow Q$	$\equiv$	$\neg P \vee Q$	
(19)	$P \Leftrightarrow Q$	$\equiv$	$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$	
(20)	$P \Leftrightarrow Q$	$\equiv$	$(P \wedge Q) \vee (\neg P \wedge \neg Q)$	

2. (a) **[1 point]** Let $f : A \to B$ be a function. Write down the definition of the preimage of a set $S \subseteq B$ with respect to f.

Solution: $f^{-1}(S) := \{a \in A : f(a) \in B\}$

(b) **[1 point]** Write down the definition of a reflexive relation on a set X.

Solution: The relation R on X is called reflexive if xRx for all $x \in X$. Equivalently, $(x, x) \in R$ for all $x \in X$.

(c) **[3 points]** Prove that if R and R' are two transitive relations on a set X then $R \cap R'$ is also a transitive relation on X.

Solution: Suppose that $(x, y) \in R \cap R'$ and $(y, z) \in R \cap R'$. Then we have $(x, y) \in R$ and $(y, z) \in R$, hence by transitivity of R we obtain $(x, z) \in R$. Similarly, from transitivity of R' we obtain $(x, z) \in R'$. From $(x, z) \in R$ and $x, z) \in R'$ it follows that $(x, z) \in R \cap R'$.

3. Set $A := \{1, 2, \ldots, 100\}$ and let $f : \mathcal{P}(\mathbb{N}) \to \mathbb{N}$ be the function defined as follows:

$$f(S) := \begin{cases} |S \cap A| & \text{if } S \cap A \neq \varnothing, \\ 5 & \text{otherwise.} \end{cases}$$

(a) **[2 points]** Compute $f(S)$ where $S = \{\varnothing, \mathbb{N}, 2\mathbb{N}\}$. You do not need to justify your answer.

Here $2\mathbb{N} := \{a \in \mathbb{N} : a \text{ is an even number}\}$.

Solution: We have $f(\varnothing) = 5$, $f(\mathbb{N}) = 100$, and $f(2\mathbb{N}) = 50$. Thus we have $f(S) = \{5, 50, 100\}$.

(b) **[3 points]** Prove or disprove the following assertion:

$$\forall_{S \in \mathcal{P}(\mathbb{N})} \big[(f(S) \neq 5) \Rightarrow (\exists_{T \subseteq S} (f(T) = 1)) \big].$$

Solution: We prove this assertion. Take any $S \subseteq \mathbb{N}$. Assume that $f(S) \neq 5$. Then by the definition of f it follows that $S \cap A \neq \varnothing$. Thus we can choose an element $a \in S \cap A$. Now consider the set $T := \{a\}$. Then $T \subseteq S$ and we have $f(T) = |T \cap A| = |\{a\}| = 1$. This completes the proof of existence of T.

4. **[2 points]** For each of the following assertions write T if it is true and write F if it is false. For each incorrect answer you will receive -.25 pts. There will be no negative points if you leave your response blank.

(a) ___ There exist two sets X and Y such that $\mathcal{P}(X) = \mathcal{P}(Y)$ but $X \neq Y$.

(b) ___ A function $f : X \to X$ is a surjection if and only if there exists another function $g : X \to X$ such that $f \circ g$ is a bijection.

(c) ___ The binary operation $A \star B := A \setminus B$ on $\mathcal{P}(\mathbb{N})$ has a left identity but not a right identity.

(d) ___ If formulas ϕ and ψ are tautologies, then $\phi \vee \neg\psi$ is also a tautology.

Solution: *F*, *T*, *F*, *T*

Solution: .5 pts for each correct answer; -.25 pts for each incorrect answer; if the total score is negative, they receive zero on this question.

5. [**4 points**] Determine the value of $a \in \mathbb{R}$ such that

$$\bigcup_{n\in\mathbb{N}} \left[\tfrac{1}{n^2+1} + 1, 2\right] = (a, 2].$$

You should justify your answer with a proof that the two sets are indeed equal.

Solution: We prove equality of both sides for $a = 1$. Take any $x \in \mathbb{R}$. There are 3 cases to consider.

- $x \leq 1$. Then we have
$$1 + \frac{1}{n^2+1} > 1 \geq x \text{ and } 2 > x,$$
hence $x \notin (1, 2]$.
- $1 < x \leq 2$. We have
$$x \in [1 + \frac{1}{1+n^2}, 2] \Leftrightarrow 1 + \frac{1}{1+n^2} \leq x \leq 2.$$
Thus, proving $x \in [1 + \frac{1}{1+n^2}, 2]$ is equivalent to proving that
$$(*) \qquad \exists_{n\in\mathbb{N}} 1 + \frac{1}{1+n^2} \leq x \leq 2.$$
Clearly the inequality $x \leq 2$ is satisfied independent of n. For the left hand side inequality,
$$\begin{aligned} 1 + \frac{1}{1+n^2} \leq x &\Leftrightarrow \frac{1}{1+n^2} \leq x - 1 \\ &\Leftrightarrow n^2 + 1 \geq \frac{1}{x-1} \quad \text{since both sides are positive} \\ &\Leftrightarrow n^2 \geq \frac{1}{x-1} - 1. \end{aligned}$$
Since natural numbers are unbounded from above, there exists $n \in \mathbb{N}$ such that the latter inequality holds. This completes the proof of $(*)$.
- $x > 2$. In this case for all $n \in \mathbb{N}$ we have
$$x > 2 \geq \frac{1}{1+n^2},$$
hence clearly $x \notin [1 + \frac{1}{1+n^2}, 2]$. Thus, we have proved that
$$\forall_{n\in\mathbb{N}} x \notin [1 + \frac{1}{1+n^2}, 2].$$
This implies that $x \notin \bigcup_{n\in\mathbb{N}} [1 + \frac{1}{1+n^2}, 2]$.